Atiyah–Macdonald, Chapter 10: Completions

Solutions to the exercises

Conventions. $\hat{A}$, $\hat{M}$ denote $\mathfrak{a}$-adic completions ($\hat{M} = \varprojlim M/\mathfrak{a}^n M$) and $\phi : M \to \hat{M}$ the natural map. From the text I use: $\operatorname{Ker}\phi = \bigcap_n \mathfrak{a}^n M$ (the intersection of all neighbourhoods of 0) and ϕ injective iff M is Hausdorff (10.1); $\varprojlim$ is left exact and is exact on surjective systems (10.2), with $\varprojlim^1 A_n = \operatorname{Coker} d^A$ and the six-term sequence $0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to \varprojlim{}^1 A_n \to \varprojlim{}^1 B_n \to \varprojlim{}^1 C_n \to 0$ coming from the snake lemma; for an exact sequence $0 \to G' \to G \to G'' \to 0$ with the induced topologies $\{G' \cap G_n\}$ and $\{pG_n\}$, the completed sequence $0 \to \hat{G}' \to \hat{G} \to \hat{G}'' \to 0$ is exact (10.3); $\hat{G} \cong \varprojlim G/G_n$ (10.4), and a ring or module is called *complete* if ϕ is an isomorphism; for A Noetherian and M finitely generated, the Artin–Rees lemma (10.9–10.11) makes the $\mathfrak{a}$-adic topology of a submodule M' the induced one; exactness of completion on finitely generated modules (10.12); $\hat{A} \otimes_A M \cong \hat{M}$ for M finitely generated (10.13); $\hat{A}$ is A-flat (10.14); $\hat{\mathfrak{a}} = \hat{A}\mathfrak{a}$ is contained in the Jacobson radical of $\hat{A}$ (10.15(iv)); the completion of a Noetherian local ring is local (10.16); Krull's theorem (10.17): $\bigcap_n \mathfrak{a}^n M = \{x \in M : (1-\alpha)x = 0 \text{ for some } \alpha \in \mathfrak{a}\}$, and 10.19; $\hat{A}$ is Noetherian (10.26). Earlier chapters: Ch. 3, Ex. 16 (faithful flatness) and Ex. 17; Ch. 2, Ex. 8; Ch. 1, Ex. 5.

This chapter has no numbered in-text exercises and 12 end-of-chapter exercises. In Exercises 9 and 10(i) the polynomial f is monic in Hensel's lemma; for 10(i) I give a Newton-iteration proof which does not need monicity (this is needed for 10(iii)).

End-of-chapter exercises

Exercise 1

> **Problem.** Let $\alpha_n : \mathbf{Z}/p\mathbf{Z} \to \mathbf{Z}/p^n\mathbf{Z}$ be the injection of abelian groups given by $\alpha_n(1) = p^{n-1}$, and let $\alpha : A \to B$ be the direct sum of all the α_n (where A is a countable direct sum of copies of $\mathbf{Z}/p\mathbf{Z}$, and B is the direct sum of the $\mathbf{Z}/p^n\mathbf{Z}$). Show that the p-adic completion of A is just A but that the completion of A for the topology induced from the p-adic topology on B is the direct *product* of the $\mathbf{Z}/p\mathbf{Z}$. Deduce that p-adic completion is *not* a right-exact functor on the category of all $\mathbf{Z}$-modules.

Solution. Write $A = \bigoplus_{k\geq 1} A_{(k)}$ with $A_{(k)} = \mathbf{Z}/p$, $B = \bigoplus_{k\geq 1} B_{(k)}$ with $B_{(k)} = \mathbf{Z}/p^k$, and $\alpha = \bigoplus \alpha_k$. α is injective since each α_k is.

p-adic completion of A. $pA = 0$, so $p^n A = 0$ for $n \geq 1$; the p-adic topology is discrete and $\hat{A} = \varprojlim A/p^n A = A$.

Induced topology. Put $A_n = \alpha^{-1}(p^n B)$ ($n \geq 0$); these form the fundamental system of the induced topology. Now $p^n B = \bigoplus_k p^n \mathbf{Z}/p^k$. For $k \leq n$, $p^n\mathbf{Z}/p^k = 0$ so $\alpha_k(A_{(k)}) \cap p^n B_{(k)} = 0$; for $k \geq n+1$, $\alpha_k(1) = p^{k-1} \in p^n\mathbf{Z}/p^k$, so $\alpha_k(A_{(k)}) \subseteq p^n B_{(k)}$. Hence

$$A_n = \bigoplus_{k>n} A_{(k)}, \qquad A/A_n \cong \bigoplus_{k\leq n} A_{(k)},$$

and the completion for this topology is $\varprojlim A/A_n = \prod_{k\geq 1} \mathbf{Z}/p$ (a coherent sequence in $\bigoplus_{k\leq n}\mathbf{Z}/p$ is the same as a sequence $(c_k)_{k\geq 1}$ of elements of $\mathbf{Z}/p$). Write A' for this completion.

Failure of right exactness. Let $C = B/\alpha(A)$ and consider the exact sequence $0 \to A \xrightarrow{\alpha} B \to C \to 0$. Give B the p-adic topology, $\alpha(A) \cong A$ the induced topology (so $\hat{A}$ there is A' by the above), and C the quotient topology $\{pB_n + \alpha A/\alpha A\}$, which is the p-adic topology of C. By 10.3,

$$0 \to A' \to \hat{B} \to \hat{C} \to 0$$

is exact. Suppose p-adic completion were right exact. Then $\hat{A} \xrightarrow{\hat{\alpha}} \hat{B} \to \hat{C} \to 0$ would be exact, where $\hat{A} = A$ (p-adic). By functoriality (10.1 ff.: $\hat{g}\circ\hat{f} = \widehat{gf}$), $\hat{\alpha}$ is the composite $A = \hat{A} \to A' \to \hat{B}$ where the first map is the completion of the continuous identity $A \to A$ (p-adic topology $\to$ induced topology), i.e. the inclusion $\bigoplus \mathbf{Z}/p \hookrightarrow \prod \mathbf{Z}/p$, and the second map is injective (10.3). Exactness would give $\operatorname{Im}\hat{\alpha} = \operatorname{Ker}(\hat{B} \to \hat{C}) = \operatorname{Im}(A' \to \hat{B})$, hence $\bigoplus \mathbf{Z}/p = \prod \mathbf{Z}/p$, false since $\bigoplus_{k\geq 1}\mathbf{Z}/p \subsetneq \prod_{k\geq 1}\mathbf{Z}/p$ (e.g. $(1,1,1,\dots)$). So p-adic completion is not right exact on all $\mathbf{Z}$-modules. □

Exercise 2

> **Problem.** In Exercise 1, let $A_n = \alpha^{-1}(p^n B)$, and consider the exact sequence $0 \to A_n \to A \to A/A_n \to 0$. Show that $\varprojlim$ is not right exact, and compute $\varprojlim^1 A_n$.

Solution. By Exercise 1, $A_n = \bigoplus_{k>n}\mathbf{Z}/p$ and $A/A_n = \bigoplus_{k\leq n}\mathbf{Z}/p$. The three systems are $\{A_n\}$ (transition maps the inclusions $A_{n+1} \hookrightarrow A_n$), $\{A\}$ (constant, transition maps the identity) and $\{A/A_n\}$ (projections), and $0 \to \{A_n\} \to \{A\} \to \{A/A_n\} \to 0$ is an exact sequence of inverse systems. We compute

$$\varprojlim A_n = \bigcap_n A_n = 0, \qquad \varprojlim A = A = \bigoplus \mathbf{Z}/p, \qquad \varprojlim A/A_n = \prod \mathbf{Z}/p,$$

and the map $\varprojlim A \to \varprojlim A/A_n$ is the inclusion $\bigoplus \mathbf{Z}/p \hookrightarrow \prod \mathbf{Z}/p$, which is not surjective. Thus $0 \to \varprojlim A_n \to \varprojlim A \to \varprojlim A/A_n$ is exact (10.2) but $\varprojlim A \to \varprojlim A/A_n$ is not onto: $\varprojlim$ is not right exact.

Computation of $\varprojlim^1 A_n$. The constant system $\{A\}$ is surjective (identity maps), so $\varprojlim^1 A = \operatorname{Coker} d^A = 0$ (see 10.2: to solve $x_n - x_{n+1} = a_n$ put $x_1 = 0$, $x_{n+1} = x_n - a_n$). The six-term sequence gives

$$0 \to \varprojlim A_n \to \varprojlim A \to \varprojlim A/A_n \to \varprojlim{}^1 A_n \to \varprojlim{}^1 A = 0,$$

so

$$\varprojlim{}^1 A_n \cong \operatorname{Coker}\Big(\bigoplus_{k\geq 1}\mathbf{Z}/p \to \prod_{k\geq 1}\mathbf{Z}/p\Big) = \Big(\prod_{k\geq 1}\mathbf{Z}/p\Big)\Big/\Big(\bigoplus_{k\geq 1}\mathbf{Z}/p\Big),$$

a non-zero (indeed uncountable) group. □

Exercise 3

Problem. Let A be a Noetherian ring, $\mathfrak{a}$ an ideal and M a finitely generated A-module. Using Krull's Theorem and Exercise 14 of Chapter 3, prove that

$$\bigcap_{n=1}^{\infty} \mathfrak{a}^n M = \bigcap_{\mathfrak{m} \supseteq \mathfrak{a}} \mathrm{Ker}(M \to M_{\mathfrak{m}}),$$

where $\mathfrak{m}$ runs over all maximal ideals containing $\mathfrak{a}$. Deduce that $\hat{M} = 0 \iff \mathrm{Supp}(M) \cap V(\mathfrak{a}) = \emptyset$ (in $\mathrm{Spec}(A)$). [The reader should think of $\hat{M}$ as the "Taylor expansion" of M transversal to the subscheme $V(\mathfrak{a})$: the above result then shows that M is determined in a neighborhood of $V(\mathfrak{a})$ by its Taylor expansion.]

Solution. Let $E = \bigcap \mathfrak{a}^n M$. By Krull's theorem (10.17), $E = \{x \in M : (1-\alpha)x = 0 \text{ for some } \alpha \in \mathfrak{a}\}$.

($\subseteq$) Let $x \in E$, $(1-\alpha)x = 0$, $\alpha \in \mathfrak{a}$, and let $\mathfrak{m} \supseteq \mathfrak{a}$ be maximal. Then $1 - \alpha \notin \mathfrak{m}$ (otherwise $1 \in \mathfrak{m}$), and $(1-\alpha)x = 0$, so $x/1 = 0$ in $M_{\mathfrak{m}}$.

($\supseteq$) Let $x \in M$ with $x/1 = 0$ in $M_{\mathfrak{m}}$ for all maximal $\mathfrak{m} \supseteq \mathfrak{a}$. Put $\mathfrak{b} = \mathrm{Ann}(x)$. For each such $\mathfrak{m}$ there is $s \notin \mathfrak{m}$ with $sx = 0$, i.e. $\mathfrak{b} \not\subseteq \mathfrak{m}$. If $\mathfrak{a} + \mathfrak{b} \neq (1)$ it would lie in a maximal ideal $\mathfrak{m}$ containing $\mathfrak{a}$ and $\mathfrak{b}$, a contradiction. So $\mathfrak{a} + \mathfrak{b} = (1)$: $1 = \alpha + b$ with $\alpha \in \mathfrak{a}$, $b \in \mathfrak{b}$, and $(1-\alpha)x = bx = 0$. Hence $x \in E$. (This is the argument of Ch. 3, Ex. 14 in disguise.)

Deduction. $\ker(M \to \hat{M}) = E$ (10.1). If $\hat{M} = 0$, then $E = M$, so by the displayed equality $x/1 = 0$ in $M_{\mathfrak{m}}$ for every $x \in M$ and every maximal $\mathfrak{m} \supseteq \mathfrak{a}$, i.e. $M_{\mathfrak{m}} = 0$ for all maximal $\mathfrak{m} \supseteq \mathfrak{a}$. Conversely if all these $M_{\mathfrak{m}} = 0$, then $E = M$, so $M = \mathfrak{a}^n M$ for all n, $M/\mathfrak{a}^n M = 0$ and $\hat{M} = 0$. Thus $\hat{M} = 0 \iff M_{\mathfrak{m}} = 0$ for all maximal $\mathfrak{m} \in V(\mathfrak{a})$.

Finally, $M_{\mathfrak{m}} = 0$ for all maximal $\mathfrak{m} \in V(\mathfrak{a}) \iff \mathrm{Supp}(M) \cap V(\mathfrak{a}) = \emptyset$: ($\Leftarrow$) maximal ideals of $V(\mathfrak{a})$ are not in the support. ($\Rightarrow$) if $\mathfrak{p} \in \mathrm{Supp}(M) \cap V(\mathfrak{a})$, choose a maximal $\mathfrak{m} \supseteq \mathfrak{p} \supseteq \mathfrak{a}$; $M_{\mathfrak{p}}$ is a localization of $M_{\mathfrak{m}}$ (Ch. 3, Ex. 3), so $M_{\mathfrak{m}} = 0 \Rightarrow M_{\mathfrak{p}} = 0$, contradicting $\mathfrak{p} \in \mathrm{Supp}\, M$. □

Exercise 4

Problem. Let A be a Noetherian ring, $\mathfrak{a}$ an ideal in A, and $\hat{A}$ the $\mathfrak{a}$-adic completion. For any $x \in A$, let $\hat{x}$ be the image of x in $\hat{A}$. Show that

$$x \text{ not a zero-divisor in } A \Rightarrow \hat{x} \text{ not a zero-divisor in } \hat{A}.$$

Does this imply that A is an integral domain $\Rightarrow$ $\hat{A}$ is an integral domain? [Apply the exactness of completion to the sequence $0 \to A \xrightarrow{x} A$.]

Solution. *First claim.* If x is not a zero-divisor, $0 \to A \xrightarrow{\mu_x} A$ is an exact sequence of finitely generated modules. By 10.12 (more precisely its left-exact part, 10.2/10.12), $0 \to \hat{A} \xrightarrow{\widehat{\mu_x}} \hat{A}$ is exact. The completion of μ_x is the continuous extension of multiplication by x, i.e. multiplication by $\hat{x}$

(both are continuous and agree on the dense image of A; equivalently, $\widehat{\mu_x} = \varprojlim(\mu_x \text{ on } A/\mathfrak{a}^n)$). So $\hat{x}$ is not a zero-divisor.

The implication is false. Let k be a field of characteristic $\neq 2$ (e.g. $\mathbf{Q}$), $B = k[x,y]$, $f = y^2 - x^2(1+x)$, $A = B/(f)$ and $\mathfrak{a} = (x,y)A$. f is irreducible in B (the polynomial $y^2 - x^2(1+x)$ over $k(x)$ has discriminant $4x^2(1+x)$, not a square in $k(x)$, so it is irreducible over $k(x)$; it is primitive in y over $k[x]$, hence irreducible in B by Gauss). So A is a Noetherian domain. By 10.13 applied to the B-module A and the ideal $(x,y) \subset B$ (whose adic topology on A is the $\mathfrak{a}$-adic one), $\hat{A} \cong \hat{B} \otimes_B A = k[[x,y]]/(f)$, since $\hat{B} = \varprojlim B/(x,y)^n = k[[x,y]]$. Let $u = \sqrt{1+x} = \sum_{n\geq 0} \binom{1/2}{n} x^n \in k[[x]]$ (here 2 is invertible in k; the coefficients $\binom{1/2}{n}$ lie in $\mathbf{Z}[1/2]$, being $\pm$ Catalan numbers divided by powers of 2), so $u^2 = 1 + x$ and

$$f = y^2 - x^2u^2 = (y - xu)(y + xu) \quad \text{in } k[[x,y]].$$

Neither factor lies in (f) (elements of (f) have order ≥ 2, while $y \mp xu$ has order 1). Hence $(y - xu)$ and $(y + xu)$ are non-zero zero-divisors of $\hat{A}$, and $\hat{A}$ is not a domain although A is (the nodal cubic: completion at the node has two branches). $\square$

Exercise 5

> **Problem.** Let A be a Noetherian ring and let $\mathfrak{a}, \mathfrak{b}$ be ideals in A. If M is any A-module, let $M^{\mathfrak{a}}$, $M^{\mathfrak{b}}$ denote its $\mathfrak{a}$-adic and $\mathfrak{b}$-adic completions respectively. If M is finitely generated, prove that $(M^{\mathfrak{a}})^{\mathfrak{b}} \cong M^{\mathfrak{a}+\mathfrak{b}}$. [Take the $\mathfrak{a}$-adic completion of the exact sequence $0 \to \mathfrak{b}^m M \to M \to M/\mathfrak{b}^m M \to 0$ and apply (10.13). Then use the isomorphism $\varprojlim_m (\varprojlim_n M/(\mathfrak{a}^n M + \mathfrak{b}^m M)) \cong \varprojlim_n M/(\mathfrak{a}^n M + \mathfrak{b}^n M)$ and the inclusions $(\mathfrak{a}+\mathfrak{b})^{2n} \subseteq \mathfrak{a}^n + \mathfrak{b}^n \subseteq (\mathfrak{a}+\mathfrak{b})^n$.]

Solution. Here $(M^{\mathfrak{a}})^{\mathfrak{b}}$ is the $\mathfrak{b}$-adic completion of the A-module $M^{\mathfrak{a}}$ (filtration $\mathfrak{b}^m M^{\mathfrak{a}}$).

Step 1. $\mathfrak{b}^m M$ and $M/\mathfrak{b}^m M$ are finitely generated, so by 10.12 the $\mathfrak{a}$-adic completion of $0 \to \mathfrak{b}^m M \to M \to M/\mathfrak{b}^m M \to 0$ is exact:

$$0 \to (\mathfrak{b}^m M)^{\mathfrak{a}} \to M^{\mathfrak{a}} \to (M/\mathfrak{b}^m M)^{\mathfrak{a}} \to 0.$$

By 10.13 and flatness of $\hat{A} = \hat{A}^{\mathfrak{a}}$ over A (10.14), $(\mathfrak{b}^m M)^{\mathfrak{a}} \cong \hat{A} \otimes \mathfrak{b}^m M \to \hat{A} \otimes M \cong M^{\mathfrak{a}}$ is injective with image the $\hat{A}$-submodule generated by the elements $1 \otimes bx$ ($b \in \mathfrak{b}^m$), i.e. $\mathfrak{b}^m(\hat{A} \otimes M) = \mathfrak{b}^m M^{\mathfrak{a}}$ (naturality of the isomorphism $\hat{A} \otimes M \cong M^{\mathfrak{a}}$). Hence

$$M^{\mathfrak{a}}/\mathfrak{b}^m M^{\mathfrak{a}} \cong (M/\mathfrak{b}^m M)^{\mathfrak{a}} = \varprojlim_n M/(\mathfrak{a}^n M + \mathfrak{b}^m M),$$

because the $\mathfrak{a}$-adic completion of $M/\mathfrak{b}^m M$ is $\varprojlim_n (M/\mathfrak{b}^m M)/\mathfrak{a}^n(M/\mathfrak{b}^m M) = \varprojlim_n M/(\mathfrak{a}^n M + \mathfrak{b}^m M)$.

Step 2. Taking $\varprojlim_m$:

$$(M^{\mathfrak{a}})^{\mathfrak{b}} = \varprojlim_m M^{\mathfrak{a}}/\mathfrak{b}^m M^{\mathfrak{a}} \cong \varprojlim_m \varprojlim_n X_{m,n}, \qquad X_{m,n} = M/(\mathfrak{a}^n M + \mathfrak{b}^m M),$$

with surjective transition maps in each index. An element of the left side is a family $(\xi_{m,n})$ with $\xi_{m,n} \in X_{m,n}$, compatible under both families of projections. The map to $\varprojlim_n X_{n,n}$, $(\xi_{m,n}) \mapsto$

$(\xi_{n,n})$, is an isomorphism: the inverse sends (η_n) to $\xi_{m,n} :=$ the image of η_k for any $k \geq \max(m,n)$ (independent of k by coherence). So $(M^{\mathfrak{a}})^{\mathfrak{b}} \cong \varprojlim_n M/(\mathfrak{a}^n + \mathfrak{b}^n)M$.

Step 3. $(\mathfrak{a}+\mathfrak{b})^{2n} \subseteq \mathfrak{a}^n + \mathfrak{b}^n$ (each monomial in $2n$ factors has $\geq n$ factors from $\mathfrak{a}$ or $\geq n$ from $\mathfrak{b}$) and $\mathfrak{a}^n+\mathfrak{b}^n \subseteq (\mathfrak{a}+\mathfrak{b})^n$. So the filtrations $((\mathfrak{a}^n+\mathfrak{b}^n)M)$ and $((\mathfrak{a}+\mathfrak{b})^n M)$ are interleaved, define the same topology, and have the same inverse limit: $\varprojlim M/(\mathfrak{a}^n+\mathfrak{b}^n)M \cong \varprojlim M/(\mathfrak{a}+\mathfrak{b})^n M = M^{\mathfrak{a}+\mathfrak{b}}$. $\square$

Exercise 6

> **Problem.** Let A be a Noetherian ring and $\mathfrak{a}$ an ideal in A. Prove that $\mathfrak{a}$ is contained in the Jacobson radical of A if and only if every maximal ideal of A is closed for the $\mathfrak{a}$-topology. (A Noetherian topological ring in which the topology is defined by an ideal contained in the Jacobson radical is called a *Zariski ring*. Examples are local rings and (by (10.15)(iv)) $\mathfrak{a}$-adic completions.)

Solution. For an ideal $\mathfrak{b}$ of A, the closure of $\mathfrak{b}$ in the $\mathfrak{a}$-topology is $\bar{\mathfrak{b}} = \bigcap_n (\mathfrak{b} + \mathfrak{a}^n)$: $x \in \bar{\mathfrak{b}}$ iff every neighbourhood $x + \mathfrak{a}^n$ meets $\mathfrak{b}$ iff $x \in \mathfrak{b} + \mathfrak{a}^n$ for all n.

($\Rightarrow$) Let $\mathfrak{a} \subseteq \mathfrak{R}$ and $\mathfrak{m}$ maximal. Then $\mathfrak{a} \subseteq \mathfrak{m}$, so $\mathfrak{a}^n \subseteq \mathfrak{m}$ and $\mathfrak{m} + \mathfrak{a}^n = \mathfrak{m}$ for all n; hence $\bar{\mathfrak{m}} = \mathfrak{m}$.

($\Leftarrow$) Suppose every maximal $\mathfrak{m}$ is closed. If $\mathfrak{a} \not\subseteq \mathfrak{m}$ for some maximal $\mathfrak{m}$, then $\mathfrak{m} + \mathfrak{a} = A$ by maximality, and $1 \in (\mathfrak{m} + \mathfrak{a})^n \subseteq \mathfrak{m} + \mathfrak{a}^n$ for all n, so $\bar{\mathfrak{m}} = A \neq \mathfrak{m}$, a contradiction. Hence $\mathfrak{a} \subseteq \mathfrak{m}$ for every maximal $\mathfrak{m}$, i.e. $\mathfrak{a} \subseteq \mathfrak{R}$. (Noetherianity is not needed.) $\square$

Exercise 7

> **Problem.** Let A be a Noetherian ring, $\mathfrak{a}$ an ideal of A, and $\hat{A}$ the $\mathfrak{a}$-adic completion. Prove that $\hat{A}$ is faithfully flat over A (Chapter 3, Exercise 16) if and only if A is a Zariski ring (for the $\mathfrak{a}$-topology). [Since $\hat{A}$ is flat over A, it is enough to show that $M \to \hat{M}$ is injective for all finitely generated $M \iff A$ is Zariski; now use (10.19) and Exercise 6.]

Solution. $\hat{A}$ is A-flat by 10.14. By Exercise 6, A is Zariski iff $\mathfrak{a} \subseteq \mathfrak{R}(A)$.

($\Rightarrow$) Suppose $\hat{A}$ is faithfully flat but $\mathfrak{a} \not\subseteq \mathfrak{m}$ for some maximal $\mathfrak{m}$. Let $M = A/\mathfrak{m} \neq 0$. Then $\mathfrak{a} + \mathfrak{m} = A$ gives $\mathfrak{a}M = M$, so $\mathfrak{a}^n M = M$ and $\hat{M} = \varprojlim M/\mathfrak{a}^n M = 0$. But by 10.13 ($M$ finitely generated) $\hat{A} \otimes_A M \cong \hat{M} = 0$, contradicting Ch. 3, Ex. 16(iv) (for a faithfully flat algebra B, $M \neq 0 \Rightarrow B \otimes M \neq 0$). Hence $\mathfrak{a} \subseteq \mathfrak{R}$. (Equivalently: faithful flatness forces $M \to \hat{M} = \hat{A} \otimes M$ to be injective for all finitely generated M, i.e. $\bigcap \mathfrak{a}^n M = 0$, which for $M = A/\mathfrak{m}$ says $\mathfrak{a} \subseteq \mathfrak{m}$.)

($\Leftarrow$) Let $\mathfrak{a} \subseteq \mathfrak{R}$. By Ch. 3, Ex. 16 ((iii)$\Rightarrow$ faithfully flat) it suffices to show $\mathfrak{m}\hat{A} \neq \hat{A}$ for each maximal $\mathfrak{m}$ of A. By 10.13, $\hat{A}/\mathfrak{m}\hat{A} \cong \hat{A} \otimes_A (A/\mathfrak{m}) \cong (A/\mathfrak{m})^\wedge$; since $\mathfrak{a} \subseteq \mathfrak{R} \subseteq \mathfrak{m}$, $\mathfrak{a}(A/\mathfrak{m}) = 0$ and the $\mathfrak{a}$-adic completion of $A/\mathfrak{m}$ is $A/\mathfrak{m} \neq 0$. Hence $\mathfrak{m}\hat{A} \neq \hat{A}$, and $\hat{A}$ is faithfully flat over A. (Alternatively: for M finitely generated $M \to \hat{M}$ is injective by 10.19, as $\mathfrak{a} \subseteq \mathfrak{R}$.) $\square$

Exercise 8

Problem. Let A be the local ring of the origin in $\mathbf{C}^n$ (i.e., the ring of all rational functions $f/g \in \mathbf{C}(z_1, \ldots, z_n)$ with $g(0) \neq 0$), let B be the ring of power series in $z_1, \ldots, z_n$ which converge in some neighborhood of the origin, and let C be the ring of formal power series in $z_1, \ldots, z_n$, so that $A \subset B \subset C$. Show that B is a local ring and that its completion for the $\mathfrak{m}$-adic topology is C. Assuming that B is Noetherian, prove that B is A-flat. [Use Chapter 3, Exercise 17, and Exercise 7 above.]

Solution. *B is local.* The evaluation $B \to \mathbf{C}$, $f \mapsto f(0)$ is a surjective ring homomorphism; its kernel $\mathfrak{m}_B$ is a maximal ideal. If $f(0) \neq 0$ then $1/f$ is holomorphic (hence given by a convergent power series) near 0, so f is a unit of B. Hence every element outside $\mathfrak{m}_B$ is a unit and B is local with maximal ideal $\mathfrak{m}_B$.

Powers of $\mathfrak{m}_B$. Clearly $\mathfrak{m}_B^k \subseteq \{f \in B : \text{all monomials of } f \text{ have degree} \geq k\}$ (the right side is an ideal containing the generators of $\mathfrak{m}_B^k$ once we know $\mathfrak{m}_B = (z_1, \ldots, z_n)B$, which is the case $k = 1$ below). Conversely let $f = \sum c_\alpha z^\alpha \in B$ have only monomials of degree $\geq k$, and let $r > 0$ with $\sum |c_\alpha| r^{|\alpha|} < \infty$ (absolute convergence at $(r, \ldots, r)$). For every α fix $\beta(\alpha) \leq \alpha$ (componentwise) with $|\beta(\alpha)| = k$. Put $f_\beta = \sum_{\beta(\alpha)=\beta} c_\alpha z^{\alpha-\beta}$. Then $\sum |c_\alpha| r^{|\alpha|-k} < \infty$, so each f_β converges on the polydisc $|z_j| \leq r$, i.e. $f_\beta \in B$, and $f = \sum_{|\beta|=k} z^\beta f_\beta \in \mathfrak{m}_B^k$ (using $k = 1$ first to see $\mathfrak{m}_B = (z_1, \ldots, z_n)B$, since $f(0) = 0$ means all monomials have degree ≥ 1, and then $\mathfrak{m}_B^k = (z^\beta : |\beta| = k)B$). Hence $\mathfrak{m}_B^k = \{f \in B : \text{all monomials have degree} \geq k\}$.

Completion. The truncation map $B \to \mathbf{C}[z]/(z)^k$ (take terms of degree $< k$) is surjective (polynomials lie in B) with kernel $\mathfrak{m}_B^k$ by the above. So $B/\mathfrak{m}_B^k \cong C/\mathfrak{m}_C^k$ compatibly in k, where $\mathfrak{m}_C = (z_1, \ldots, z_n)C$, hence $\hat{B} = \varprojlim B/\mathfrak{m}_B^k = \varprojlim C/\mathfrak{m}_C^k = C$.

Flatness of $A \to B$. $A = S^{-1}\mathbf{C}[z]$ with $S = \mathbf{C}[z] \setminus (z_1, \ldots, z_n)$ is Noetherian and local with maximal ideal $\mathfrak{m}_A = A \cap \mathfrak{m}_B$, and $A/\mathfrak{m}_A^k \cong \mathbf{C}[z]/(z)^k \cong C/\mathfrak{m}_C^k$ (localization at the maximal ideal (z) does not change $\mathbf{C}[z]/(z)^k$). So the $\mathfrak{m}_A$-adic completion of A is $\hat{A} = C$, with natural map $A \to C$ the Taylor expansion at 0; it equals the composite $A \to B \to \hat{B} = C$. Now:

- $A \to \hat{A} = C$ is flat by 10.14 (A Noetherian).
- $B \to \hat{B} = C$ is faithfully flat by Exercise 7: B is Noetherian (assumed) and local, so $\mathfrak{m}_B \subseteq \mathfrak{R}(B)$ and B is a Zariski ring.

By Ch. 3, Ex. 17 applied to $A \to B \to C$ (the composite $A \to C$ is flat, $B \to C$ faithfully flat), $A \to B$ is flat. □

Exercise 9

Problem. Let A be a local ring, $\mathfrak{m}$ its maximal ideal. Assume that A is $\mathfrak{m}$-adically complete. For any polynomial $f(x) \in A[x]$, let $\bar{f}(x) \in (A/\mathfrak{m})[x]$ denote its reduction mod $\mathfrak{m}$. Prove *Hensel's lemma*: if $f(x)$ is monic of degree n and if there exist coprime monic polynomials $\bar{g}(x), \bar{h}(x) \in (A/\mathfrak{m})[x]$ of degrees $r, n-r$ with $\bar{f}(x) = \bar{g}(x)\bar{h}(x)$, then we can lift $\bar{g}(x), \bar{h}(x)$ back to monic polynomials $g(x), h(x) \in A[x]$ such that $f(x) = g(x)h(x)$. [Assume inductively that we have constructed $g_k(x), h_k(x) \in A[x]$ such that $g_k(x)h_k(x) - f(x) \in \mathfrak{m}^k[x]$. Then use the fact that since $\bar{g}(x)$ and $\bar{h}(x)$ are coprime we can find $\bar{a}_p(x), \bar{b}_p(x)$, of degrees $\leq n-r$, r respectively, such that $x^p = \bar{a}_p(x)\bar{g}_k(x) + \bar{b}_p(x)\bar{h}_k(x)$, where p is any integer such that $1 \leq p \leq n$. Finally, use the completeness of A to show that the sequences $g_k(x), h_k(x)$ converge to the required $g(x), h(x)$.]

Solution. Let $\kappa = A/\mathfrak{m}$.

Linear algebra over κ. Since $\bar{g}, \bar{h}$ are coprime of degrees r and $n-r$, the κ-linear map $(a,b) \mapsto a\bar{g}+b\bar{h}$ from $\{\deg a < n-r\}\times\{\deg b < r\}$ (dimension n) to $\{\deg < n\}$ (dimension n) is injective: if $a\bar{g} = -b\bar{h}$ then $\bar{h} \mid a\bar{g}$, so $\bar{h} \mid a$ by coprimality, whence $a = 0$ as $\deg a < \deg \bar{h}$, and then $b = 0$. So it is bijective: for each $0 \le p \le n-1$ there are $\bar{a}_p, \bar{b}_p$ with $\deg \bar{a}_p < n-r$, $\deg \bar{b}_p < r$ and $x^p = \bar{a}_p\bar{g} + \bar{b}_p\bar{h}$.

Inductive construction. Choose monic lifts $g_1, h_1 \in A[x]$ of $\bar{g}, \bar{h}$ of degrees r, $n-r$; then $g_1h_1 - f \in \mathfrak{m}[x]$ (the reductions agree). Suppose g_k, h_k monic of degrees $r, n-r$, reducing to $\bar{g}, \bar{h}$, with $e := f - g_kh_k \in \mathfrak{m}^k[x]$. Both f and g_kh_k are monic of degree n, so $\deg e \le n-1$; write $e = \sum_{p=0}^{n-1} e_p x^p$, $e_p \in \mathfrak{m}^k$. Lift $\bar{a}_p, \bar{b}_p$ to $a_p, b_p \in A[x]$ with $\deg a_p < n-r$, $\deg b_p < r$ (the reductions of g_k, h_k are $\bar{g}, \bar{h}$), so that $x^p - a_pg_k - b_ph_k \in \mathfrak{m}[x]$. Put

$$g_{k+1} = g_k + \sum_p e_p b_p, \qquad h_{k+1} = h_k + \sum_p e_p a_p.$$

These are monic of degrees r, $n-r$ (we added polynomials of degrees $< r$, $< n-r$), congruent to g_k, h_k mod $\mathfrak{m}^k[x]$, and

$$g_{k+1}h_{k+1} = g_kh_k + \sum_p e_p(a_pg_k + b_ph_k) + \Big(\sum e_pb_p\Big)\Big(\sum e_pa_p\Big).$$

Now $\sum_p e_p(a_pg_k + b_ph_k) = \sum e_px^p - \sum e_p(x^p - a_pg_k - b_ph_k) \in e + \mathfrak{m}^{k+1}[x]$, and the last term lies in $\mathfrak{m}^{2k}[x] \subseteq \mathfrak{m}^{k+1}[x]$. So $g_{k+1}h_{k+1} - f = g_kh_k + e - f + (\mathfrak{m}^{k+1}[x]) = 0 + \mathfrak{m}^{k+1}[x]$.

Convergence. The coefficients of g_k (resp. h_k) form Cauchy sequences, since $g_{k+1} - g_k \in \mathfrak{m}^k[x]$. As A is complete, they converge to $g, h \in A[x]$ monic of degrees r, $n-r$, with $g-g_k, h-h_k \in \mathfrak{m}^k[x]$ (the $\mathfrak{m}^k$ are closed), and $\bar{g}, \bar{h}$ are the given reductions. Then $gh - f \equiv g_kh_k - f \equiv 0 \pmod{\mathfrak{m}^k[x]}$ for all k, so the coefficients of $gh - f$ lie in $\bigcap_k \mathfrak{m}^k = 0$ (complete rings are Hausdorff). Thus $f = gh$. □

Exercise 10

Problem.

(i) With the notation of Exercise 9, deduce from Hensel's lemma that if $\bar{f}(x)$ has a simple root $\alpha \in A/\mathfrak{m}$, then $f(x)$ has a simple root $a \in A$ such that $\alpha = a \bmod \mathfrak{m}$.

(ii) Show that 2 is a square in the ring of 7-adic integers.

(iii) Let $f(x,y) \in k[x,y]$, where k is a field, and assume that $f(0,y)$ has $y = a_0$ as a simple root. Prove that there exists a formal power series $y(x) = \sum_{n=0}^{\infty} a_nx^n$ such that $f(x, y(x)) = 0$. (This gives the "analytic branch" of the curve $f = 0$ through the point $(0, a_0)$.)

Solution. (i) *Monic case, from Hensel.* If f is monic and $\bar{f} = (x-\alpha)\bar{h}$ with $\bar{h}(\alpha) \neq 0$ (simple root), then $x - \alpha$ is irreducible and does not divide $\bar{h}$, so $x - \alpha$ and $\bar{h}$ are coprime monic polynomials of degrees $1, n-1$. Hensel gives $f = (x-a)h$ with $a \equiv \alpha$ and $h \in A[x]$ monic; a is a root, and $h(a) \equiv \bar{h}(\alpha) \neq 0 \pmod{\mathfrak{m}}$, so $h(a)$ is a unit, in particular non-zero: the root is simple.

General f (no monicity), by Newton iteration. Let $\bar{f}(\alpha) = 0$, $\bar{f}'(\alpha) \neq 0$ (this is what "simple root" means: $\bar{f} = (x-\alpha)\bar{q}$, $\bar{q}(\alpha) = \bar{f}'(\alpha) \neq 0$). Pick a lift $a_1 \in A$ of α; then $f(a_1) \in \mathfrak{m}$ and $f'(a_1) \notin \mathfrak{m}$ is a unit of the local ring A. Define recursively $a_{k+1} = a_k - f(a_k)/f'(a_k)$, and check by induction: $f(a_k) \in \mathfrak{m}^k$, $a_{k+1} - a_k \in \mathfrak{m}^k$ (so $f'(a_k) \equiv f'(a_1) \bmod \mathfrak{m}$ is a unit). Taylor's formula in $A[t]$ gives $f(a+t) = f(a) + f'(a)t + t^2 c$ with $c \in A$; with $a = a_k$, $t = -f(a_k)/f'(a_k) \in \mathfrak{m}^k$:

$$f(a_{k+1}) = t^2 c \in \mathfrak{m}^{2k} \subseteq \mathfrak{m}^{k+1}.$$

The sequence (a_k) is Cauchy, so converges (completeness) to $a \in A$ with $a \equiv a_1 \equiv \alpha \pmod{\mathfrak{m}}$, and $f(a) \equiv f(a_k) \equiv 0 \bmod \mathfrak{m}^k$ for all k, so $f(a) = 0$. Also $f'(a) \equiv f'(a_1) \not\equiv 0 \bmod \mathfrak{m}$, so writing $f = (x-a)q$ we get $q(a) = f'(a)$ a unit: a is a simple root.

(ii) Let $A = \mathbf{Z}_7$, the 7-adic integers: the completion of $\mathbf{Z}$ (equivalently of the local ring $\mathbf{Z}_{(7)}$) at (7); it is local with maximal ideal $7\mathbf{Z}_7$ and residue field $\mathbf{Z}/7$ (10.16), and complete (10.5). Take $f = x^2 - 2$. Modulo 7, $\bar{f}(3) = 9 - 2 = 0$ and $\bar{f}'(3) = 6 \neq 0$, so 3 is a simple root of $\bar{f}$. By (i) there is $a \in \mathbf{Z}_7$ with $a^2 = 2$.

(iii) Take $A = k[[x]]$, a complete local ring with maximal ideal (x) and residue field k. View $f(x,y) \in k[x][y] \subseteq A[y]$, a polynomial in the variable y with coefficients in A. Its reduction mod (x) is $\bar{f}(y) = f(0,y)$, which has the simple root $a_0 \in k$. By (i) (general form) there is $y(x) \in A = k[[x]]$ with $f(x, y(x)) = 0$ and $y(0) = a_0$. □

Exercise 11

> **Problem.** Show that the converse of (10.26) is false, even if we assume that A is local and that $\hat{A}$ is a finitely-generated A-module. [Take A to be the ring of germs of C^∞ functions of x at $x = 0$, and use Borel's Theorem that every power series occurs as the Taylor expansion of some C^∞ function.]

Solution. (10.26: A Noetherian $\Rightarrow$ $\hat{A}$ Noetherian.) Let A be the ring of germs at 0 of real-valued C^∞ functions on $\mathbf{R}$.

A is local with $\mathfrak{m} = (x)$. If $f(0) \neq 0$, then $1/f$ is C^∞ near 0, so f is a unit of A; hence $\mathfrak{m} = \{f : f(0) = 0\}$ is the unique maximal ideal. By Taylor's formula with integral remainder, for $f \in A$ and $n \geq 1$,

$$f(x) = \sum_{j<n} \frac{f^{(j)}(0)}{j!} x^j + x^n g_n(x), \qquad g_n(x) = \int_0^1 \frac{(1-t)^{n-1}}{(n-1)!} f^{(n)}(tx)\, dt \in A.$$

So $f \in x^n A \iff f^{(j)}(0) = 0$ for all $j < n$ ("⇒" by Leibniz). In particular $\mathfrak{m} = xA$ and $\mathfrak{m}^n = x^n A = \{f : f^{(j)}(0) = 0,\ j < n\}$.

The completion. The Taylor map $T : A \to \mathbf{R}[[x]]$ is a ring homomorphism, with $T(f) \bmod x^n$ the Taylor polynomial of degree $< n$. By the above, $A/\mathfrak{m}^n \cong \mathbf{R}[x]/(x^n)$ via the Taylor polynomial (onto, as polynomials are in A). Hence $\hat{A} = \varprojlim A/\mathfrak{m}^n = \mathbf{R}[[x]]$, and the natural map $A \to \hat{A}$ is T. By Borel's theorem T is surjective, so $\hat{A} = A \cdot 1$ is a finitely generated A-module (and a Noetherian ring: $\mathbf{R}[[x]]$ is, by 10.27).

A is not Noetherian. Suppose it were. Then Krull's theorem (10.17) applied to $M = A$ and $\mathfrak{a} = \mathfrak{m}$ gives $\bigcap_n \mathfrak{m}^n = \{f : (1-\alpha)f = 0 \text{ for some } \alpha \in \mathfrak{m}\}$. But $1 - \alpha$ is a unit (since $\alpha \in \mathfrak{m}$, A

local), so the right side is 0. However $\bigcap_n \mathfrak{m}^n = \operatorname{Ker} T$ is the ideal of flat germs, and contains the non-zero germ e^{-1/x^2} (extended by 0 at 0), all of whose derivatives vanish at 0. This contradiction shows A is not Noetherian, although A is local and $\hat{A}$ is a Noetherian, finitely generated A-module. □

Exercise 12

> **Problem.** If A is Noetherian, then $A[[x_1, \ldots, x_n]]$ is a faithfully flat A-algebra. [Express $A \to A[[x_1, \ldots, x_n]]$ as a composition of flat extensions, and use Exercise 5(v) of Chapter 1.]

Solution. Write $A \to P = A[x_1, \ldots, x_n] \to \hat{P} = A[[x_1, \ldots, x_n]]$.

Flatness. $P = \bigoplus_\alpha Ax^\alpha$ is a free A-module, hence flat. P is Noetherian (Hilbert basis theorem), and $\hat{P}$ is its $(x_1, \ldots, x_n)$-adic completion: $P/(x)^k$ is the set of polynomials of degree $< k$, so $\varprojlim P/(x)^k = A[[x_1, \ldots, x_n]]$. By 10.14, $\hat{P}$ is P-flat. A composite of flat homomorphisms is flat (Ch. 2, Ex. 8(ii)), so $A[[x_1, \ldots, x_n]]$ is a flat A-algebra.

Faithfulness. Let $\mathfrak{p}$ be a prime of A and $\varepsilon : A[[x_1, \ldots, x_n]] \to A$ the evaluation $x_i \mapsto 0$ (constant term). Then $\varepsilon^{-1}(\mathfrak{p})$ is a prime of $A[[x_1, \ldots, x_n]]$ (the quotient embeds in $A/\mathfrak{p}$: it is $\cong A/\mathfrak{p}$, a domain), and its contraction to A is $\varepsilon^{-1}(\mathfrak{p}) \cap A = \mathfrak{p}$ (for $a \in A$, $\varepsilon(a) = a$). (For $n = 1$ this is Ch. 1, Ex. 5(v); the general case is the same, or iterate $A[[x_1, \ldots, x_n]] = A[[x_1, \ldots, x_{n-1}]][[x_n]]$.) Thus $\operatorname{Spec} A[[x_1, \ldots, x_n]] \to \operatorname{Spec} A$ is surjective, and a flat algebra with this property is faithfully flat (Ch. 3, Ex. 16: condition (ii) is one of the equivalent conditions defining faithful flatness). □